

Practical no. 13

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Class :- CO3K – B

Subject :- OOP

Aim :- Define all the classes to represent hierarchy and define functions to retrieve individual information as and when required.

Program :-

```
#include<iostream.h>
#include<conio.h>
class Staff{
public:
int code;
char name[30];
};
class Teacher:public Staff{
public:
char subject[40];
char publication[50];
void t_accept(){
cout<<"\nEnter Details of Teacher :- ";
cout<<"\nEnter your Code :- ";
cin>>code;
cout<<"\nEnter your Name :- ";
cin>>name;
cout<<"\nEnter your Subject :- ";
cin>>subject;
cout<<"\nEnter your Publication :- ";
cin>>publication;
}
void t_display(){
cout<<"Details of Teacher :- ";
cout<<"\nCode :- "<<code;
cout<<"\nName :- "<<name;
cout<<"\nSubject :- "<<subject;
cout<<"\nPublication :- "<<publication;
}
};
class Officer: public Staff{
public:
char grade;
void o_accept(){
cout<<"\nEnter details of Officer :- ";
```

```

cout<<"\nEnter your Code :- ";
cin>>code;
cout<<"\nEnter your Name :- ";
cin>>name;
cout<<"\nEnter your Grade :- ";
cin>>grade;
}
void o_display(){
cout<<"\nDetails of Officer :- ";
cout<<"\nCode :- "<<code;
cout<<"\nName :- "<<name;
cout<<"\nGrade :- "<<grade;
}
};
class Typist: public Staff{
public:
int speed;
};
class Regular:public Typist{
public:
int salary;
void r_accept(){
cout<<"\nEnter details of Regular Typist :- ";
cout<<"\nEnter your Code :- ";
cin>>code;
cout<<"\nEnter your Name :- ";
cin>>name;
cout<<"\nEnter your Typing Speed :- ";
cin>>speed;
cout<<"\nEnter your Salary :- ";
cin>>salary;
}
void r_display(){
cout<<"\nDetails of Regular Typist :- ";
cout<<"\nCode :- "<<code;
cout<<"\nName :- "<<name;
cout<<"\nTyping Speed :- "<<speed;
cout<<"\nSalary :- "<<salary;
}
};
class Casual: public Typist{
public:
int daily_wages;
void c_accept(){
cout<<"\nEnter details of Casual Typist :- ";

```

```

cout<<"\nEnter your Code :- ";
cin>>code;
cout<<"\nEnter your Name :- ";
cin>>name;
cout<<"\nEnter your Typing Speed :- ";
cin>>speed;
cout<<"\nEnter your Daily Wages :- ";
cin>>daily_wages;
}
void c_display(){
cout<<"\nDetails of Casual Typist :- ";
cout<<"\nCode :- "<<code;
cout<<"\nName :- "<<name;
cout<<"\nTyping Speed :- "<<speed;
cout<<"\nDaily Wages :- "<<daily_wages;
}
};
void main(){
Teacher t;
Officer o;
Regular r;
Casual c;
clrscr();
t.t_accept();
t.t_display();
getch();
clrscr();
o.o_accept();
o.o_display();
getch();
clrscr();
r.r_accept();
r.r_display();
getch();
clrscr();
c.c_accept();
c.c_display();
getch();
}

```

Output :-

```
NeuTroN DOS-C++ 0.77, Cpu speed: max 100%  
  
Enter Details of Teacher :-  
Enter your Code :- 805  
  
Enter your Name :- Danish  
  
Enter your Subject :- OOP  
  
Enter your Publication :- MSBTE  
Details of Teacher :-  
Code :- 805  
Name :- Danish  
Subject :- OOP  
Publication :- MSBTE
```

```
NeuTroN DOS-C++ 0.77, Cpu speed: r  
  
Enter details of Officer :-  
Enter your Code :- 805  
  
Enter your Name :- Danish  
  
Enter your Grade :- A  
  
Details of Officer :-  
Code :- 805  
Name :- Danish  
Grade :- A_
```

```
NeuTroN DOS-C++ 0.77, Cpu speed: max 100%  
  
Enter details of Regular Typist :-  
Enter your Code :- 805  
  
Enter your Name :- Danish  
  
Enter your Typing Speed :- 30  
  
Enter your Salary :- 10000  
Details of Regular Typist :-  
Code :- 805  
Name :- Danish  
Typing Speed :- 30  
Salary :- 10000_
```

```
NeuTroN DOS-C++ 0.77, Cpu speed: max 100%  
  
Enter details of Casual Typist :-  
Enter your Code :- 805  
  
Enter your Name :- Danish  
  
Enter your Typing Speed :- 20  
  
Enter your Daily Wages :- 500  
  
Details of Casual Typist :-  
Code :- 805  
Name :- Danish  
Typing Speed :- 20  
Daily Wages :- 500
```

Aim :- WAP in C++ to implement given class hierarchy.

Program :-

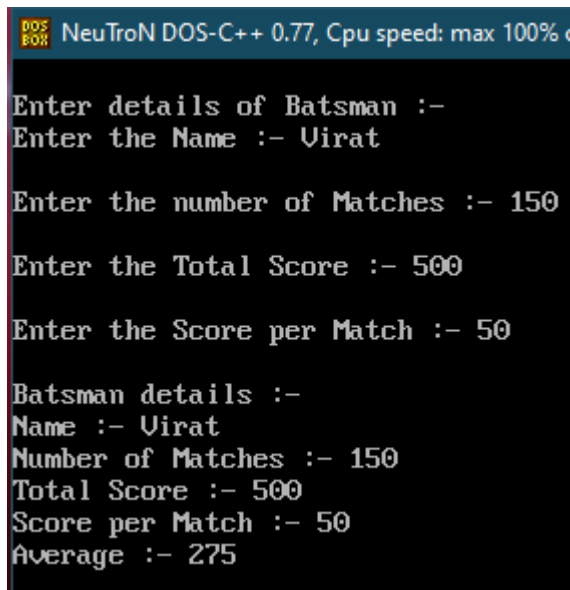
```
#include<iostream.h>
#include<conio.h>
class Player{
public:
char name[50];
int matches;
};
class Batsman:public Player{
public:
int total_score;
int per_match_score;
int average;
void accept(){
cout<<"\nEnter details of Batsman :- ";
cout<<"\nEnter the Name :- ";
cin>>name;
cout<<"\nEnter the number of Matches :- ";
cin>>matches;
cout<<"\nEnter the Total Score :- ";
cin>>total_score;
cout<<"\nEnter the Score per Match :- ";
cin>>per_match_score;
}
void calculate_Average(){
int avg;
avg=(total_score+per_match_score)/2;
average=avg;
}
void display(){
cout<<"\nBatsman details :- ";
cout<<"\nName :- "<<name;
cout<<"\nNumber of Matches :- "<<matches;
cout<<"\nTotal Score :- "<<total_score;
cout<<"\nScore per Match :- "<<per_match_score;
cout<<"\nAverage :- "<<average;
}
};
class Bowler:public Player{
public:
int no_of_wickets;
void b_accept(){
cout<<"\nEnter details of Bowler :- ";
cout<<"\nEnter the Name :- ";
cin>>name;
cout<<"\nEnter the Number of Matches :- ";
cin>>matches;
cout<<"\nEnter the Number of Wickets :- ";
```

```

cin>>no_of_wickets;
}
void b_display(){
cout<<"\nBowler details :- ";
cout<<"\nName :- "<<name;
cout<<"\nNumber of Matches :- "<<matches;
cout<<"\nNumber of Wickets :- "<<no_of_wickets;
}
};
void main(){
Batsman b1;
Bowler b2;
clrscr();
b1.accept();
b1.calculate_Average();
b1.display();
getch();
clrscr();
b2.b_accept();
b2.b_display();
getch();
}

```

Output :-



```

DOS BOX NeuTroN DOS-C++ 0.77, Cpu speed: max 100%
Enter details of Batsman :-
Enter the Name :- Virat

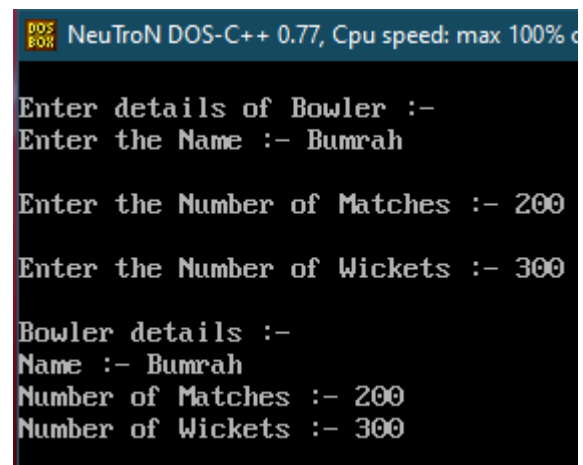
Enter the number of Matches :- 150

Enter the Total Score :- 500

Enter the Score per Match :- 50

Batsman details :-
Name :- Virat
Number of Matches :- 150
Total Score :- 500
Score per Match :- 50
Average :- 275

```



```

DOS BOX NeuTroN DOS-C++ 0.77, Cpu speed: max 100%
Enter details of Bowler :-
Enter the Name :- Bumrah

Enter the Number of Matches :- 200

Enter the Number of Wickets :- 300

Bowler details :-
Name :- Bumrah
Number of Matches :- 200
Number of Wickets :- 300

```